

Multi-Purpose Reefs – A Decade of Applications

Shaw Mead¹ and Jose Borrero¹

¹ASR Limited, Marine Consulting and Research, Raglan, New Zealand; s.mead@asrltd.co.nz

Abstract

Multi-Purpose or artificial Reefs (MPRs) have been a topic at the Australasian Coasts and Ports Conference since 1997. Seven MPRs have now been constructed at locations around the world with differing levels of success. This paper provides a brief update on the performance of these structures, summarizing the large volume of monitoring data available for many of these projects. When the first MPR was constructed in 1999, it was a novel and innovative project. More than a decade on and with several more MPRs in the water, a great deal has been learned about the functionality of and potential applications for these structures. This includes shoreline protection and beach response, ecological enhancement, habitat restoration and improvement of recreational amenities. We discuss several projects with respect to the above topics as well as some of the lessons learned, such as the impact of scale on amenity, in the light of future MPR developments.

Keywords: coastal structures, sediment transport, performance, recreational facilities, design

1 Introduction

Offshore submerged structures for coastal protection [14] and artificial reefs for enhancing local marine biodiversity and/or attracting sea life [3] have been utilized for over a century. Indeed, the first known use of artificial reef structures for habitat enhancement dates back to 2000 BC on the west coast of India [9]. However, it was not until Grigg's 1969 Surfer Magazine article that MPRs to combine coastal protection, eco-enhancement and surfing were first proposed.

Through the 1970's and 1980's several more articles on the subject of MPR's were put forward, including the seminal works of James 'Kimo' Walker who pioneered the area of surfing science and developed the first methods for quantification of a surfing break. Design studies for artificial reefs primarily for surfing were undertaken for the construction of the world's first artificial surfing reef built in Perth in 1998, as well as for artificial reefs in California and the Queensland Gold Coast MPR in Australia.

In the past decade, the understanding of many aspects of MPR performance, as well as the analytical and numerical tools to design such structures has greatly increased, particularly in terms of coastal protection, as well as ecological and surfing enhancement.



Figure 1. Locations of MPR's around the world. From left to right, El Segundo (California, USA), Boscombe (UK), Kovalam (India), Cables (Perth, Australia), Narrownneck (Gold Coast, Australia), Opunake (New Zealand), Mount Maunganui (NZ).

It is worth noting that the acronym MPR has been used more frequently in recent years to describe artificial surfing reefs. However, of the 7 artificial reefs constructed, only 2 can be considered 'true' MPR's that

encompass coastal protection, surfing and ecological enhancement; these are the Narrownneck Reef, in Australia and the Kovalam Reef in India (Figure 1).

This paper considers the performance of MPR's in three primary areas of functionality: coastal protection, ecological enhancement and surfing enhancement. We also discuss some of the lessons learned and applied as understanding of these structures has increased.

2 Ecological Enhancement

In many respects, the ecological enhancement aspects of MPR's can be considered an artefact of these structures. In ecological terms, the principles are simple and well known; hard stable substrates result in greater biodiversity and species abundance than mobile sandy substrates [3]. Thus, the placement of MPR's offshore and underwater results in a local increase in biodiversity [6].

Colonization of MPR's has been observed and recorded to occur very quickly. Within just a few days, pioneer species such as hydroids can be observed attached to the surface of the structure. Within months to years, it can be difficult for the untrained eye to distinguish the MPR from a natural reef.

Of interest in this area of MPR technology is the type and abundance of different species as a function of different construction materials. This topic has been considered in great detail in many different parts of the world for reefs designed specifically for ecological enhancement (e.g. see [3]). However, MPR's differ from these traditional artificial reefs in that they are located within or just outside of the surf zone rather than in deep water.

Several investigations of individual reefs have catalogued the species inhabiting MPR's and compared them to natural rocky reefs nearby. These studies have found that while biodiversity indices are similar, there are distinct differences in species composition on MPR's constructed of sand-filled containers (SFC's) in comparison to the natural rocky reefs, i.e. there are comparatively higher numbers of soft-bodied encrusting species (e.g. sponges, acidians, etc – Figure 2) and lower numbers of hard-bodied species (e.g. barnacles, tubeworms, etc.) on the MPR's [6].



Figure 2. Ascidians (seasquirts) and encrusting algae inhabiting the SFC's of Boscombe Reef in the UK 6 months after placement on the seabed, which immediately became an attraction for the local crab population.

A recent study comparing the colonization rates and biodiversity on a range of different substrates and a range of different geotextile materials found that non-woven geotextiles (the material used to fabricate SFC's in 5 of the 7 MPR's built to date) had higher colonization rates and abundances than concrete, rock, steel and other geotextile materials [5]. This kind of comparison with respect to potential ecological enhancement has become a consideration in the design of MPR's in recent years, and has recently been applied to coastal protection in California where studies suggest that there are lower numbers of seabirds present where rock walls are used for coastal protection as compared to beach nourishment (M. McGinnis, pers. comm.).

3 Surfing Enhancement

The surfing aspect of MPR's has without doubt drawn the most attention and caught the imagination of the surfing community and the general public and is often accompanied by a great deal of media speculation. As a result, there has been large discrepancy between expectations for the surfing waves on a particular MPR compared to reality [4]. Even though there has been some success at creating quality surfing waves, the general public perception is that artificial surfing reefs have mostly failed to create good waves for surfing.

Shand [17] presents a considered discussion on why there has been such a difference between the public expectation and the reality of the surfing aspects of MPR's. He considers the range of aspects associated with the development of an MPR including the community, design and construction. He concludes that the single largest limitation to building a successful artificial reef in terms of surfing is the issue of scale. He notes that to in order optimise surfing, MPR structures must be much larger in volume than most built to date to not be dependent on the joint occurrence of a small range of wave and tide conditions. While we are in agreement with Shand's [17] philosophical review of incorporating surfing into MPR's, here we summarize the available information on the surfing performance of the 7 MPR's built to date.

3.1 Results of Surfing Enhancement on MPR's

3.1.1 Cable Station, Perth, Western Australia

Cable Station was designed at the University of Western Australia [10]. The 5,000 m³ MPR was

constructed in 1998 of limestone rock placed from a barge. The site was considered an 'almost' surfing break prior to construction, with the design developed toward creating a 50 m long right hand breaker (i.e. when the surfer catches the wave he/she turns right to surf on the peeling wave face).

Following construction, the public reaction to the surfing enhancement of the structure was generally negative, and set the pattern for almost every subsequent MPR built, i.e. that the reef wasn't producing good surfing waves and that it was a failure. Similar to MPR projects that would follow Cable Station, media-driven expectations were not met when people visited the Reef and were not greeted by the "Perfect Wave".

Indeed, there is a belief among the general public, the media and even a portion of the surfing population that MPR's 'make' waves in a manner similar to a wave pool, rather than 'breaking' waves like a natural reef and thereby being reliant on a particular set of conditions for it to work properly. In addition, as described by Shand [17], due to the relatively small structural volume of a typical MPR as compared to a natural surfing break, that set of conditions is often fairly restricted, resulting in a fickle surfing break that does not consistently produce good quality waves but may produce occasional periods of good waves when the right set of conditions come together.

It is readily apparent that Cable Station is not the best location for an MPR designed primarily for surfing. This is due to it being sheltered by Rottnest Island and natural fringing reefs that result in a small wave climate with a narrow swell window. Nevertheless, Pattiarachi [13] reported an increase from 5-7 surfable days at this site prior to reef construction to 142 surfable days following construction.

While the common perception is that Cable Station does not regularly produce high-quality surfing waves, perhaps the question with respect to what is a successful project could be better framed through a comparison between the before and after surfing amenity. Using this framework to convey such information to the mainstream media both prior to and following construction may have resulted in a different public perception [17].

3.1.2 Narroreel Reef, Gold Coast, Australia

The Narroreel Reef is the first ever MPR designed primarily for coastal protection while attempting to incorporate surfing enhancement. Similar to Cable Station, the surfing aspects of the reef were considered to not live up to the expectations built up prior to the project [8].

Although Narroreel Reef is the largest yet built (some 70,000 m³ – [8]), volume is still an issue with respect to creating a new surfing break, in this case for two different reasons. Firstly the expectation that the Reef would produce waves similar to the world-class Burleigh Heads and Kirra Points to the south on the Gold Coast. Although the surfing parameters at these 2 world-class breaks were applied to the design of the Narroreel Reef [12], the preconditioning and sheltering that the large headlands afford the 2 natural breaks could not be

replicated for both practical and logistical reasons as these natural headlands represent several **million** cubic meters of material.

Secondly, the large volume of the Narrowneck Reef required a cost-effective construction process which utilized a split-hull dredge to fill and then drop the containers into place [8]. The down side of this methodology was the difficulty in exact container positioning and the subsequent non-uniform structure. While this has detracted from the surfing amenity and reduced the surfability of the waves, there is no doubt that the surfing potential has been increased on the reef and surrounding beach breaks. Indeed, surfing rides of up to 60 s long have been recorded by surfers starting on the reef and connecting to adjacent sand bars [4]. While the Narrowneck Reef has not performed as speculated prior to construction, and the construction method has reduced the surfing amenity, there has been some success in this area, and if more realistic expectations for the project had been established and conveyed to the media, then perhaps the outcome would not have been different in terms of surfing.

3.1.3 Pratte's Reef, El Segundo, California

Pratte's Reef was built following a landmark court case where construction of a shore perpendicular groyne led to the demise of existing surfing breaks. Unfortunately, this project was doomed from the outset due to the scale problem described above. Settlement funds were used largely for permitting and legal needs leaving very little for design and construction. As a result, the smallest ever MPR (1,350 m³) was built very close to the shore and did not provide any surfing amenity.

El Segundo Reef was removed in 2010, costing more than twice the US\$300,000 it cost to build.

3.1.4 Mount Maunganui Reef, New Zealand

A 5-year consent for construction was granted in 1998 to build the Mount Maunganui Reef. The 5-year permit was based on the application for a 'Research' project through the University of Waikato.

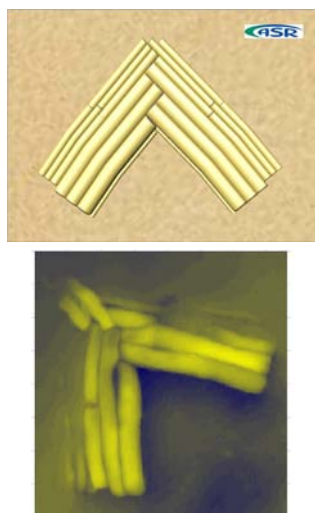


Figure 3. Comparison between the design 6,500 m³ (top) versus as-built survey 2,800 m³ (bottom) of the Mount Reef. The 2 containers end to end on the left of the picture are the replacements for the damaged and removed 60 x 6 m container.

Construction began in November 2005. From the onset of construction, the project was plagued by delays and cost over-runs, as well as a range of construction faults such as ruptured SFC's, unsealed ports leading to loss of sand, and plant failure. This resulted in a protracted construction period and the 6,500 m³ structure was never fully completed to specification (Figure 2).

In late October 2006, the right-hand side was completed close to design specification. At this stage, the reef produced very good surfing waves which broke as designed in a hollow, tubing manner (Figure 4). However, during construction, one of the largest containers was damaged beyond repair and was removed (Figure 3).

Through the summer of 2007, there were occasional reports of good quality surfing waves breaking on the Mount Reef (as evidenced by the large amount of material posted on-line from this time). However, it was some 9 months before replacement containers were put into the 60 m long x 6 m wide gap in the reef (Figure 3), and in the interim an adjacent container slumped into the gap resulting in a ~1 m lower crest height at the offshore end. In addition to the loss of sand in lower containers there is now only 2,800 m³ of the 6,500 m³ design volume currently in place (Figure 3).



Figure 4. Surfing on the Mount Reef, late summer 2007.

As a result of the smaller volume and lower crest height than designed, the Mount Reef now breaks during only a very confined set of conditions. As with the MPR's described above, it is considered a failure in terms of surfing. Also similar is the public perception of the project as 'over-promising and under-delivering', with pre-construction media claiming the "Pipeline was coming to Mount Maunganui" and an expectation that "perfect" surfing waves would be breaking on the Reef every day. Along with the lessons learnt with respect to the pitfalls of a protracted construction time, the Mount Reef is another example where realistic expectations and a framework to convey this to the media could have reduced public dismay.

3.1.5 Opunake Reef, New Zealand

Construction of the 4,800 m³ Opunake Reef commenced in early 2006, and like the Mount Reef was plagued with construction problems from the beginning. The reef location inside Opunake Bay is favourable in that the predominant winds from the northwest are mostly offshore and the waves are pre-conditioned prior to reaching to the reef site. However the reef remains unfinished with the last construction work occurring in May 2008.

At present some 1,400 m³ of reef volume is required to bring the structure up to the design crest height, and

there has been no commitment to raise the additional funds for completion. Although there was no fixed price for construction, the construction problems which have mostly been due to difficult working conditions on this very exposed coast, have lead to protracted construction time and costs approaching twice the original cost estimates.

Even though incomplete, and despite the confidence provided by the quality of the waves on the inside section of the break (which is the closest to completion), similar to previous reefs with respect to surfing enhancement, it is widely considered a failure. In this case, while public expectations are part of the reason for how the Opunake Reef is currently viewed, accurate up-to-date costing and contingency for the unforeseen (i.e. a surety that there were sufficient funds to complete the project [17]) could have led to a complete reef and positive outcome for all involved.

3.1.6 Boscombe Reef, England

The 12,800 m³ Boscombe Reef, built as part of the Boscombe Chime redevelopment, was completed in late September 2009 [11]. Bad weather and a late start in the first construction season (northern hemisphere summer of 2008) resulted in significant construction delays and a political field-day for the local media. Similar to the Mount Reef and Opunake Reef, it has been subject to claims of cost over-runs and delays, and although the latter are warranted, the Boscombe Reef was built with a fixed price contract.

Numerous bathymetry surveys and shoreline profiles were conducted before, during and after construction. These data have been used to quantify the shoreline response to compare the as-built structure to the design. In addition, an MSc study included video data capture from above the cliff above the reef and from the water to monitor surfability.

From these data it is clear that Boscombe Reef:

- Was built to design, and;
- The surfing waves on the reef break as expected for the design waves.

Boscombe Reef has a following of local surfers, and considering its location half way up the English Channel, provides regular surfing waves (<http://www.facebook.com/group.php?gid=136933823008480&ref=ts#!/boscombeboardriders>). In addition, the Boscombe Chime project has won awards (e.g. The British Chartered Surveyors Award for Urban Redevelopment 2010), changed the whole scene at this area of the waterfront and is supported by the local business owners.

Even so, despite these positive results supported by comprehensive data, the Boscombe Reef has been branded a failure by the local media. This is due in part to the results of surfing quality monitoring undertaken by Plymouth University and made available in May 2010. While the study results (downloadable on the internet at: http://www.bournemouthcho.co.uk/news/features/surfr_reef/surfreesstories/8170744.Surf_reef_it_s_only_4_out_of_10_says_independent_investigator/) indicate that the reef met 7 of the 11 criteria, the local media mis-

reported this figure as '4 out of 10' and branded the reef as an outright failure. Furthermore, only 9 performance criteria were agreed upon by the Council and the reef builders, making the final reef performance score 7/9.

Boscombe Reef is a good example of what at present could be considered the futility of recognition of a successful MPR in terms of surfing enhancement. A fair proportion of the perception is based on extended lead in time for construction of this project, with the first investigations undertaken some 10 years prior to construction. As a result there has been almost a decade of media speculation and today ludicrous statements such as "the Reef will generate 13-foot waves" are still commonly cited.

Again the points described by Shand [17] in the area of public perception may have led to quite the opposite result at Boscombe.

3.1.7 Kovalam Reef, India

The 4,000 m³ Kovalam Reef was completed in late February 2010. Kovalam Reef is only the second MPR built with coastal protection as its primary purpose with, surfing amenity and ecological enhancement as secondary considerations (i.e. an MPR).

While primarily for coastal protection, the surfing amenity at Kovalam has been enhanced greatly, even though the structure is small in volume. This has been achieved because:

- The Lighthouse end of Kovalam Beach was an 'almost' break to begin with;
- The site is dominated by clean swells and light winds during the non-monsoon season, and;
- Orientation of SFC's was such that construction tolerances could be relaxed without loss of surfing characteristics.

The Kovalam Reef has been adopted by the Indian Surfing Association and is themed on the Ministry of Tourism's website. However, being somewhat off the beaten track for travelling surfers, has so far had relatively little mainstream media attention.

4 Coastal Protection/Beach Response

Of the 7 MPR's constructed to date, 4 of them have had some kind of monitoring with respect to coastal protection.

4.1 Results of Beach Response due to MPR

4.1.1 Narrowneck Reef, Gold Coast, Australia

The primary function of the Narrowneck Reef is coastal protection. Much of the drive to build an MPR at this location was due to a cost-benefit analysis [16].

The Narrowneck Reef is part of the Gold Coast City Council's Northern Gold Coast Beach Protection Strategy, which includes renourishment and dune planting and won the Queensland Environment Award in 2000. Some 12 years after construction, monitoring via the ARGUS remote camera network has proven conclusively that the Reef has widened the beach in the lee of the structure by some 30-50 m and influences the beach over a distance of approximately 2 km [1] (Figure 5).

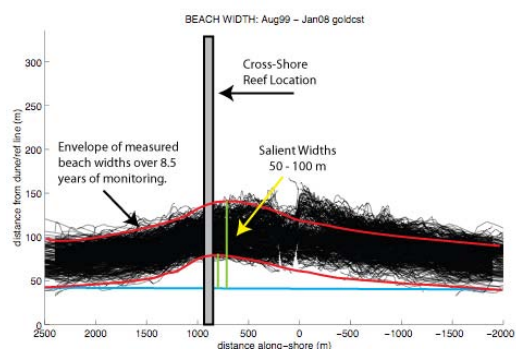


Figure 5. Shoreline position in the vicinity of the Narrowneck Reef between Aug 1999 and Jan 2008.

4.1.2 Mount Maunganui, New Zealand

The Mount Reef was designed primarily for surfing enhancement. It was constructed some 50 m closer to shore than design due to the limits of nearshore disposal grounds for material dredged from the Port of Tauranga. As a result, during large wave events, the reef is inside the surf-zone, which is counter-productive for coastal protection [15]. However, the morphological response to the structure has been studied in detail. Indeed, more than 30 Ph.D., MSc and peer-reviewed papers have focused on the morphological and ecological impacts of the structure.

Weppe [18] undertook an odd/even analysis of beach response, similar to that undertaken for the Narrowneck Reef. His results suggested that the Mount Reef caused a widening of the beach of some 25 m and a positive influence along some 150 m of beach to the southeast of the structure.

4.1.3 Boscombe Reef, England



Figure 6. Aerial view of Poole Bay at Boscombe Pier. The Boscombe Reef can be seen in the centre of the image along with the salient response in its lee stretching back to the west and the groynes spaced at approximately every 180 m.

Seventeen bathymetry surveys with complimentary beach profile surveys on 3 occasions have been recorded for Boscombe Reef. As with the Mount Reef, the primary purpose for this structure was surfing enhancement. However, Poole Bay incorporates some 60 hardwood groynes to combat coastal erosion (Figure 6). Even so, areas of Poole Bay require renourishment every 10-12 years to recharge the beach between the groynes. This is required due to the predominance of across-shore erosion driven by short-period waves generated in the English Channel. While the net direction of sediment transport is to the east, much of this occurs in the surf zone during storm events, well

beyond the ends of the groynes. Thus, sediment is first dragged offshore and then moved eastward.

Offshore detached structures are very useful for combating across-shore sediment transport, which has been confirmed in Poole Bay with the construction of Boscombe Reef. Within 6 months of construction the beach in the lee and to the west (i.e. asymmetric) of Boscombe Reef had accreted to a maximum additional width of some 60 m along more than 500 m of beach (Figure 6). The potential further utilization of MPR's in Poole Bay has been recently discussed.

4.1.4 Kovalam Reef, India

Kovalam Reef was completed in February 2010 near the end of the non-monsoon season. The primary aim was to widen and maintain the southern end of Kovalam Beach and protect the land-based infrastructure (promenade) at this popular tourist location.

Weekly photographic monitoring from the lighthouse viewing platform has provided suitable data for time-series analysis of rectified photographs. Analysis of these data suggest that during the monsoon season, the beach has remained some 10-15 m wider in comparison to pre-reef conditions and up to 30 m wider during the non-monsoon (Figure 7).



Figure 7. Kovalam Beach shoreline response during monsoon. The red line is the position of the beach pre-reef; the yellow line the beach position after. This comparison is during similar wave conditions and time of the year.

5 Summary: Lessons Learned and Future Directions

As discussed above, ecological enhancement will occur on offshore submerged structures, even when located in the surf zone, and while there are opportunities to increase this response and target specific species, this is not discussed further here.

In terms of coastal response, the results have been very positive, especially when the MPR's have been designed with this as the primary objective. There are several reasons that could lead to an increase in the application of MPR's for coastal protection in the future.

In the past decade, there has been a marked change in the attitudes previously adopted for climate change response, with 'Managed Advance' now featuring regularly as an option. Similarly, there are an increasing number of socio-economic studies that have focused on the value of a beach, which point to the tremendous asset that a wide sandy beach can be to

the local community and tourism (e.g. [7]). This is further fueled by the modern desire for natural character on the coast rather than an obvious engineered environment. In addition, there is a growing concern with respect to the sustainability of renourishment in many parts of the world. Renourishment in conjunction with structures and dune stabilization is an option that is now regularly being considered: MPR's have obvious potential in this area.

There is no doubt that with respect to surfing amenity, the perception is that MPR's have not delivered. While there has been some success in this area, the influence of persistent negative media reports has most likely stymied the development of further MPR's primarily for surfing. Indeed, a well-known and highly regarded surfing magazine, *The Surfer's Journal*, recently published an article that even questions the ecological benefits and coastal protection aspects of MPR's. It is therefore likely that the near future of MPR's primarily for surfing will ride on the success of surfing amenity associated with MPR's built primarily for coastal protection and more balanced media articles on the realities of existing MPR's (e.g. S. Nettle, Swellnet.com.au Tuesday 19th April 2011).

Many of the lessons learned in terms of the surfing benefits of MPR's are discussed in Shand [17], and include:

- The impacts of scale on the consistency of the surfing break – larger structures are required to create consistent surfing breaks;
- Managing the public perception from the outset of project (it is notable that the majority of MPR's built so far were first started more than 10 years prior to construction), and;
- Ensuring that funds are available to complete construction including all contingencies.

The science and understanding is now established and widely available for the design of MPR's for surfing enhancement and is backed up with data from existing projects (e.g. [11,12]). However, much of the actual failure (in comparison to media speculation) in terms of surfing enhancement has been due to the difficulty of converting the designs to structures on the seabed. In this area there have been several important lessons learned, and include:

1. Structures have to be built in as short a time period as possible. Protracted construction can lead to local changes in the seabed and failure/displacement of construction units that will negatively impact on the surfing amenity.
2. Small structures result in a reduced amenity due to varying wave period, and varying tide levels [2].
3. Construction tolerances can be relaxed by applying simple procedures such as orienting construction units perpendicular to the incoming wave crest [2].
4. Different construction materials should be considered for construction.

The next decade will provide interesting developments, with a likely swing back towards the primary benefits of coastal protection and a better understanding of the likely results of surfing enhancement potentially leading

to a better communication of what can be expected from MPR's. Although there is a significant amount of skepticism, there is still a great deal of interest in MPR's, and there have been measurable positive results in the areas of coastal protection and ecological enhancement. Application of the lessons learned in the areas of construction and clearer communication of the expected surfing results could lead to increased application of MPR's especially when sea level rise and managed advance are part of the equation.

6 References

- [1] Blacka, M.J., Anderson, D.J. and Mallen Lopez, L., 2008. Analysis of Shoreline Variability, Seasonality and Erosion / Accretion Trends: February 2008 - July 2008. Report 18: Northern GC Coastal Imaging System. WRL Report 2008/27.
- [2] Borrero and Mead, 2010. Borth Reef: Surfing Assessment Study. Report for Ceredigion Council, September, 2010.
- [3] Bulletin of Marine Science 1994. 55(2-3):268-275.
- [4] Corbett, B., L. A. Jackson and T. Evans, 2010a.. Narrowneck Reef Monitoring and Lessons – 10 years on. 7th Int. Surfing Reef Sym., Sydney March 19, 2010.
- [5] Corbett, B., L. A. Jackson, S. Restall and T. Evans, 2010b. Comparison of Geosynthetic Materials as Substrates on Coastal Structures – Gold Coast (Australia) and Arabian Gulf. Proc's of the 32nd ICCE, Shanghai, China.
- [6] Edwards, R. A., and S. D. A. Smith, 2005. Subtidal Assemblages Associated with a Geotextile Reef in South-East Queensland, Australia. *Mar. and Fr Res*, 2005, 56, 133-142.
- [7] Houston, J. R., 2002. The Economic Value of Our Beaches. US Army Engineer Res. and Development Centre.
- [8] Jackson, L. A., B. Corbett, R. Tomlinson, J. McGrath and G. Stuart, 2007. Narrowneck Reef: Review of 7 Years of Monitoring Results. *Shore and Beach*, 75(4):67-79.
- [9] Kurian, J., 1995. Collective action for common property resource rejuvenation: the case of people's artificial reefs in Kerala State, India. *Human Organisation* 54: 160-168
- [10] Lyons, M., 1992. Design of an Artificial Surfing Reef. BEng, Dept of Civil and Enviro. Eng., Uni. of WA Australia.
- [11] Mead S.T, C. Blenkinsopp, J. C. Borrero and A. Moores, 2010. Design and Construction of the Boscombe Multi-Purpose Reef. Proc's of 32nd ICCE. Shanghai, China.
- [12] Mead, S. T., 2000. Incorporating High-Quality Surfing Breaks into Multi-Purpose Reefs. PhD in Coastal Oceanography and Surfing Reefs thesis. Uni. of Waikato.
- [13] Pattiarachi 2007. The Cables Artificial Surfing Reef, Western Australia. *Shore and Beach* 75(4)80-92
- [14] Pilarczyk, K. W., 2003. Design of low-crested (submerged) structures – an overview. 6th Int. Conf. on Coastal and Port Eng. in Developing Countries, Sri Lanka.
- [15] Ranasinghe, R., Turner, I.L. and Symonds, G 2006. Shoreline response to multi-functional artificial surfing reefs: A numerical and physical modelling study. *C. Eng.* 53: 589-611.
- [16] Raybould, M. and T. Mules, 1998. Northern Gold Coast Beach Protection Strategy: A Benefit-Cost Analysis. Report prepared for the Gold Coast City Council, February 1998.
- [17] Shand, T. D., 2011. Making Waves? A Philosophical Review of Artificial Surfing Reef Projects. Submitted to *Shore and Beach*.
- [18] Weppe, S., T. Healy, B. Scarfe and D. Immenga, 2009. Shoreline response to an offshore submerged multifunction reef at Mount Maunganui, New Zealand. Proc's of the 19th Australasian Coasts and Ports Conference.